

Combinatorics PhD Exam, August 2017

1. Let a_n denote the number of strings of length n of the letters A, B, C, and D such that the letter A appears an odd number of times.

- (a) Find a closed formula for a_n .
- (b) Find the exponential generating function for the sequence $\{a_n\}$. (using no summation signs)

2. The set $[n] \times [n]$ is partially ordered by the relation $(a, b) \preceq (c, d)$ which holds when $a \leq c$ and $b \leq d$. Find, with proof, the length of a maximum chain and the length of a maximum antichain.

What is the size of a minimum chain decomposition of this poset. (Recall that a chain decomposition of a poset is a partition of its elements into disjoint chains.)

3. This problem has two parts.

- (a) Prove that, for a simple graph G with at least 5 vertices, at least one of G or its complement has a cycle.
- (b) Suppose that you color the edges of K_n using 2 colors. Show that there exists a monochromatic spanning tree.

4. A parity check matrix H of a binary linear code C can be defined as a generator matrix of the dual code $C^\perp$.

Show that C is the null space of the transpose H^T , where multiplication by H^T is on the right, i.e., cH^T .

Show that the minimum distance of the code equals the cardinality of a minimum dependent set of columns of H .

If a generator matrix of a binary linear code C is

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix},$$

then show that $(c_1, c_2, c_3, c_4, c_5)$ is a codeword if and only if (modulo 2)

$$\begin{aligned} c_1 + c_2 + c_3 &= 0 \\ c_2 + c_4 &= 0 \\ c_1 + c_5 &= 0. \end{aligned}$$

What is the minimum distance of this code?

5. Let $g(n)$ be the number of permutations of length n in which each cycle is of even length. Find the exponential generating function of the sequence $g(n)$, where $n = 0, 1, 2, \dots$
6. Let t_n be the total number of cycles of all permutations of length n . So $t_1 = 1$, $t_2 = 3$, and $t_3 = 11$. Find an explicit formula for the numbers t_n . Your answer can contain one summation sign.
7. Prove that the language $\{a^{n^2} : n \in \mathbb{N}\}$ is not regular.
8. Let β be a permutation of length k . Prove that there are precisely $k^2 + 1$ permutations of length $k + 1$ containing β .